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Inventory replenishment agent

Every day, for three SKUs, this agent forecasts demand with EWMA, sets safety stock from its own forecast error, compares inventory position with an order-up-to level over the lead time plus one review day, and then orders, holds, or defers. Deferring is priced: the agent waits only when the expected shortage cost of waiting is below the holding cost it saves. It also watches its own forecast for bias and logs an ALERT when it adapts.

Run it with **Runtime > Run all**. The first cell clones the repository, and everything else comes from the package in `replenish/`. After the one-page design document, the sections follow the assignment tasks T1 to T6, then the evaluation and trade-offs, the audit of the course starter notebook, the test suite, and the scaling note.

In [1]:
```
# Setup: works in Colab and locally, and re-running it changes nothing.
import importlib.util, os, subprocess, sys

REPO_URL = "https://github.com/minchree/inventory-replenishment-agent.git"
IN_COLAB = "google.colab" in sys.modules
if IN_COLAB:
    target = "/content/inventory-replenishment-agent"
    if not os.path.isdir(target):
        subprocess.run(["git", "clone", "--depth", "1", REPO_URL, target], check=True)
    os.chdir(target)
elif os.path.basename(os.getcwd()) == "notebooks":
    os.chdir("..")
if os.getcwd() not in sys.path:
    sys.path.insert(0, os.getcwd())

# Colab already has numpy, pandas, scipy and matplotlib; install only what's missing.
missing = [m for m in ("hypothesis", "pytest") if importlib.util.find_spec(m) is None]
if missing:
    subprocess.run([sys.executable, "-m", "pip", "install", "-q", *missing], check=True)

import matplotlib, numpy, pandas, scipy, hypothesis, pytest
print("Python", sys.version.split()[0], "| in Colab:", IN_COLAB, "| cwd:", os.getcwd())
for mod in (numpy, pandas, scipy, matplotlib, hypothesis, pytest):
    print(f"  {mod.__name__:<11} {mod.__version__}")
```

```
Python 3.13.18 | in Colab: True | cwd: /content/inventory-replenishment-agent
  numpy       2.4.6
  pandas      2.2.3
  scipy       1.16.3
  matplotlib  3.10.0
  hypothesis  6.170.14
  pytest      8.4.2
```

In [2]:
```
# Knobs. Service levels and lead times live in data/params.csv.
ALPHA = None         # EWMA smoothing constant; None picks it per SKU on the warm-up, e.g. 0.3 fixes it
MC_SEEDS = 200       # demand paths per policy (held-out seeds 5000+); lower it for a faster run
SWEEP_SEEDS = 100    # demand paths per service level in the sweep

import json
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from IPython.display import Image, Markdown, display

from replenish import report
from replenish.config import WARMUP_DAYS
from replenish.data import load_dataset, read_sales
from replenish.evaluate import HEADLINE, POLICIES, diagnostics, evaluate_all, metrics_table, run_dataset

pd.set_option("display.width", 200)
pd.set_option("display.max_columns", 30)
pd.set_option("display.float_format", lambda x: f"{x:,.3f}")
```

In [3]:
```
# The one-page design document: goal, inputs and outputs, guardrails, metrics, ethics and trust
display(Markdown(open("docs/DESIGN_DOCUMENT.md", encoding="utf-8").read()))
```

Design document

Goal

The agent keeps three SKUs in stock at the lowest total cost. Each day it decides, per SKU, whether to place a purchase order and how big, aiming to meet each SKU's service level while minimizing holding, stockout and ordering costs.

Inputs and outputs

Inputs, read fresh at every daily review:

- daily demand history (`sales.csv` , aggregated per SKU per day)
- on-hand stock, open POs and backorders, starting from `inventory.csv`
- per-SKU parameters from `params.csv` : lead time, minimum order quantity, service level, unit, holding and stockout costs
- the agent's own EWMA forecast and its recent forecast errors

Outputs, per SKU per day: one action (ORDER with a quantity, HOLD, or DEFER) and a log line giving the numbers behind it. An ALERT line appears when the agent adjusts its own forecast.

Guardrails

- Every PO is at least `min_order_qty` .
- Lead time is respected. Decisions use inventory position (on hand + on order − backorders) over L + 1 days, so stock already in transit is never ordered twice.
- Safety stock comes from `service_level` and the agent's measured forecast error.
- No single order exceeds 30 days of forecast demand unless the minimum order quantity requires it. This stops a bad data point or a broken forecast from triggering a huge PO.
- Waiting is allowed only when its expected stockout cost is below the holding cost it saves, and only if the service level it accepts stays within 2 points of the target. This is where holding cost is traded against stockout cost, one decision at a time.

Metrics

- Stockouts: days with any unit short, and the number of units short.
- Fill rate: units served from stock on the day they were demanded, divided by units demanded.
- Total cost: holding (per unit per day on end-of-day stock) + stockout (once per unit short) + ordering (per PO; 0 here, 25 in a second run). Purchase spend is reported separately.

Ethics and trust

A stockout breaks a promise to a customer, and excess stock ties up money that someone else has to answer for. The agent balances service reliability against cost control explicitly and shows its work. Every action is logged with the forecast, safety stock, inventory position and the cost comparison behind it, so a buyer can audit any PO, disagree with it, and see exactly which number to change. When the configured service level and the cost parameters disagree, the agent reports it (CSL*) rather than overriding a target a person set. Its tracking signal flags when its own forecast has gone wrong, so problems surface in the log instead of in empty shelves.

T1. Load and aggregate demand per day per SKU

`data/sales.csv` is at order-line grain: a SKU-day can have up to four rows, and a day with no demand has no row at all. Aggregation sums the lines per day and SKU, then reindexes to the full calendar so missing days read as zero (`replenish/data.py`).

In [4]:
```
sales = read_sales("data/sales.csv")
print(f"sales.csv: {len(sales)} order lines, {sales['date'].nunique()} dates, SKUs {sorted(sales['sku'].unique())}")
print("most lines on one SKU-day:", sales.groupby(["date", "sku"]).size().max())
display(sales.head(6))

ds = load_dataset("data")
print("\nparams.csv"); display(pd.read_csv("data/params.csv"))
print("inventory.csv"); display(pd.read_csv("data/inventory.csv"))
print(f"daily demand table: {ds.demand.shape[0]} days x {ds.demand.shape[1]} SKUs; "
      f"zero-demand days per SKU: {(ds.demand == 0).sum().to_dict()}")
display(ds.demand.head(8))
```

```
sales.csv: 690 order lines, 120 dates, SKUs ['SKU-A', 'SKU-B', 'SKU-C']
most lines on one SKU-day: 4
```

	date	sku	qty_sold
0	2025-01-01	SKU-A	10
1	2025-01-01	SKU-A	3
2	2025-01-01	SKU-A	27
3	2025-01-01	SKU-B	6
4	2025-01-02	SKU-A	32
5	2025-01-02	SKU-A	18

params.csv

	sku	unit_cost	holding_cost_per_day	stockout_cost	lead_time_days	min_order_qty	service_level
0	SKU-A	4.000	0.020	1.500	3	60	0.950
1	SKU-B	25.000	0.100	25.000	7	120	0.980
2	SKU-C	80.000	0.400	40.000	10	24	0.980

inventory.csv

	sku	opening_stock
0	SKU-A	200
1	SKU-B	90
2	SKU-C	30

daily demand table: 120 days x 3 SKUs; zero-demand days per SKU: {'SKU-A': 0, 'SKU-B': 0, 'SKU-C': 77}

sku	SKU-A	SKU-B	SKU-C
date			
2025-01-01	40	6	0
2025-01-02	50	15	0
2025-01-03	48	7	0
2025-01-04	39	14	0
2025-01-05	33	10	0
2025-01-06	24	15	0
2025-01-07	33	13	0
2025-01-08	48	8	0

T2. EWMA forecast per SKU

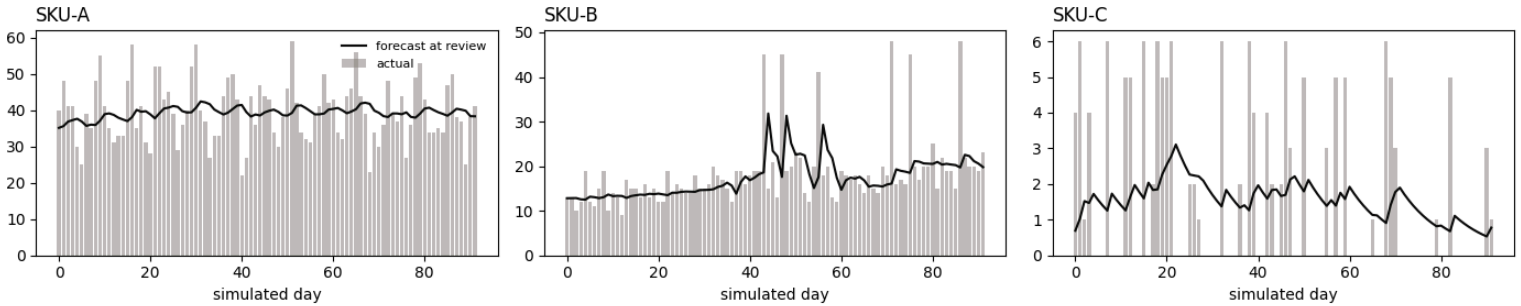
The first 28 days are warm-up. The EWMA level starts at the mean of the first week and runs through days 7 to 27, so every warm-up error is one step ahead. alpha is picked from 0.1 to 0.5 by the lowest one-step MSE. Each SKU is also classified by how often and how evenly it sells (Syntetos-Boylan ADI and CV^2), which decides whether the reflect step can trust its tracking signal.

In [5]:
```
runs = run_dataset(ds, {**POLICIES[HEADLINE], "alpha": ALPHA})
rows = []
for sku, run in runs.items():
    agent = run.policy
    rows.append({"sku": sku, "chosen alpha": agent.memory.base_alpha,
                 **{f"MSE a={a}": m for a, m in agent.alpha_mse.items()},
                 "demand class": agent.demand_class.label, "ADI": agent.demand_class.adi,
                 "CV2": agent.demand_class.cv2, "reflect active": agent.reflect_active})
display(pd.DataFrame(rows).set_index("sku"))

fig, axes = plt.subplots(1, 3, figsize=(14, 3), sharey=False)
for ax, (sku, run) in zip(axes, runs.items()):
    ax.bar(range(len(run.decisions)), run.ledger["demand"], color="#898781", alpha=0.55, label="actual")
    ax.plot([r.forecast for r in run.decisions], color="#0b0b0b", lw=1.5, label="forecast at review")
    ax.set_title(sku, loc="left"); ax.set_xlabel("simulated day")
```

```
axes[0].legend(frameon=False, fontsize=8)
plt.tight_layout(); plt.show()
```

sku	chosen alpha	MSE a=0.1	MSE a=0.2	MSE a=0.3	MSE a=0.4	MSE a=0.5	demand class	ADI	CV2	reflect active
SKU-A	0.100	111.251	114.015	116.164	116.514	114.984	smooth	1.000	0.070	True
SKU-B	0.100	7.042	7.370	7.852	8.378	8.946	smooth	1.000	0.051	True
SKU-C	0.100	3.497	3.730	4.031	4.371	4.749	intermittent	4.667	0.313	False



T3. Safety stock from forecast error and service level

With L the lead time and one review day, the protection interval is P = L + 1. Then:

- sigma_1 = RMSE of the last 28 one-step forecast errors (floored at max(0.5, 10% of the forecast))
- sigma_P = sigma_1 × √P, mu_P = forecast × P
- SS = z × sigma_P with z from `service_level`, and S = mu_P + SS

The last columns check the service level against the cost parameters. CSL* = 1 − h·T/p, where T is the order-cycle length. It is report-only. The predicted fill rate is what the normal model expects at this target; the achieved fill rate is what the simulation delivered.

In [6]:
```
diag = diagnostics(runs)
display(pd.DataFrame(diag).T[["demand_class", "alpha", "lead_time_days", "service_level", "z", "sigma1_start",
                              "sigma_P_start", "mu_P_start", "ss_start", "S_start", "mean_order", "cycle_days",
                              "csl_star", "verdict", "predicted_fill_rate", "achieved_fill_rate"]])
```

	demand_class	alpha	lead_time_days	service_level	z	sigma1_start	sigma_P_start	mu_P_start	ss_start	S_start	mean_order	cycle_days	csl_star	verdict	predicted_fill_rate	achieved_fill_rate
SKU-A	smooth	0.100	3	0.950	1.645	10.548	21.095	140.551	34.698	175.249	60.167	1.517	0.980	costs favour more protection	0.994	0.999
SKU-B	smooth	0.100	7	0.980	2.054	2.654	7.506	102.520	15.415	117.934	123.200	6.699	0.973	aligned	0.999	0.960
SKU-C	intermittent	0.100	10	0.980	2.054	1.870	6.203	7.623	12.738	20.361	24.000	15.333	0.847	costs favour less protection	0.998	0.951

T4. The daily decision: order, hold or defer

At each review the agent compares inventory position (on hand + on order − backorders) with S.

- **HOLD** when IP ≥ S.
- Below S, the order would be Q = max(MOQ, ⌈S − IP⌉). Before placing it, the agent prices waiting one day. Waiting saves h × Q of holding cost and risks dES extra units short on day t+L (from the normal loss function), at p per unit. It **DEFERs** only if waiting is cheaper and the cycle service level it accepts stays within 2 points of the target. Otherwise it **ORDERs**.
- **ALERT** lines in the T6 log come from the reflect step, when the tracking signal moves beyond 3 standard deviations of what an unbiased forecast would produce.

Three real decisions from this run, with the numbers behind them:

In [7]:
```
recs = [r for run in runs.values() for r in run.decisions]
examples = [next(r for r in recs if r.action == "HOLD"),
            next(r for r in recs if r.action == "ORDER" and r.moq_binds),
            next(r for r in recs if r.action == "DEFER")]
cols = ["day", "date", "sku", "action", "qty", "ip", "forecast", "sigma1", "mu_P", "ss", "S", "candidate",
        "csl_wait", "d_es", "wait_cost", "hold_saved", "floor"]
```

```
display(pd.DataFrame([{c: getattr(r, c) for c in cols} for r in examples]).set_index("day"))
for r in examples:
    print(report.log_lines(r)[-1])
```

day	date	sku	action	qty	ip	forecast	sigma1	mu_P	ss	S	candidate	csl_wait	d_es	wait_cost	hold_saved	floor
28	2025-01-29	SKU-A	HOLD	0	200.000	35.138	10.548	140.551	34.698	175.249	0	NaN	NaN	NaN	NaN	NaN
29	2025-01-30	SKU-A	ORDER	60	160.000	35.624	10.357	142.496	34.072	176.567	60	0.801	2.290	3.436	1.200	0.930
33	2025-02-03	SKU-A	DEFER	0	180.000	36.883	10.005	147.532	32.913	180.446	60	0.948	0.441	0.661	1.200	0.930

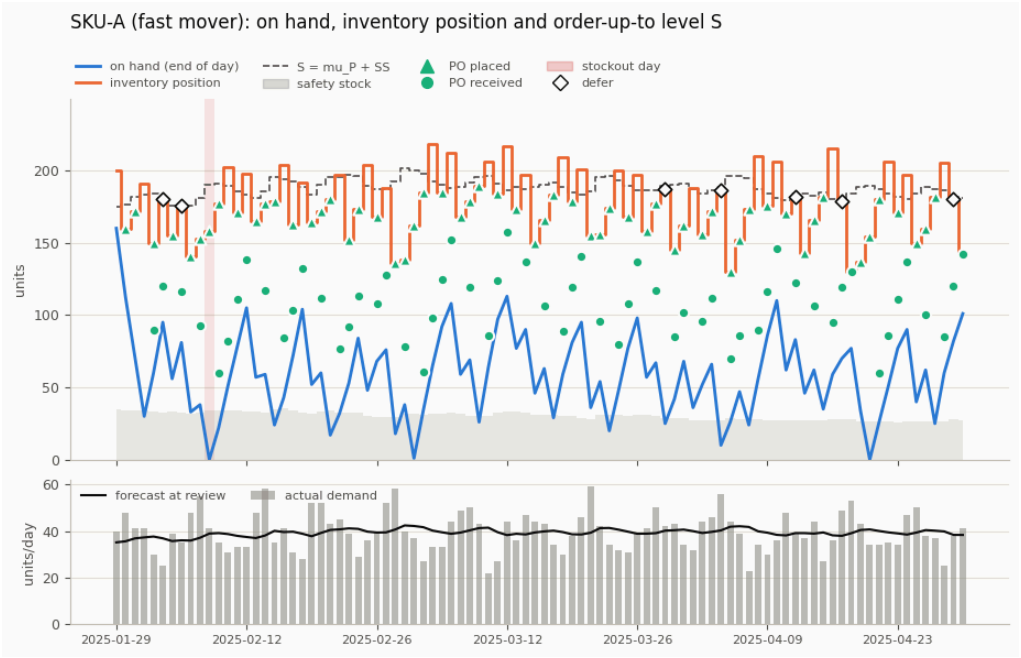
```
d028 2025-01-29 SKU-A  HOLD     0  IP=200 >= S=175.2 (mu_P=141 + SS=34.7) | 5.7 days of cover
d029 2025-01-30 SKU-A  ORDER   60  IP=160 < S=176.6 (mu_P=142 + SS=34.1, z=1.64, sigma_P=20.7) | gap 17, MOQ 60 binds | wait: +2.29u short ($3.44) >= $1.20 holding saved -> order
d033 2025-02-03 SKU-A  DEFER    0  IP=180 < S=180.4 (mu_P=148 + SS=32.9, z=1.64, sigma_P=20.0) | gap 1, MOQ 60 binds | wait: +0.44u short ($0.66) < $1.20 holding saved, CSL_wait 0.948 >= floor 0.930 -> defer
```

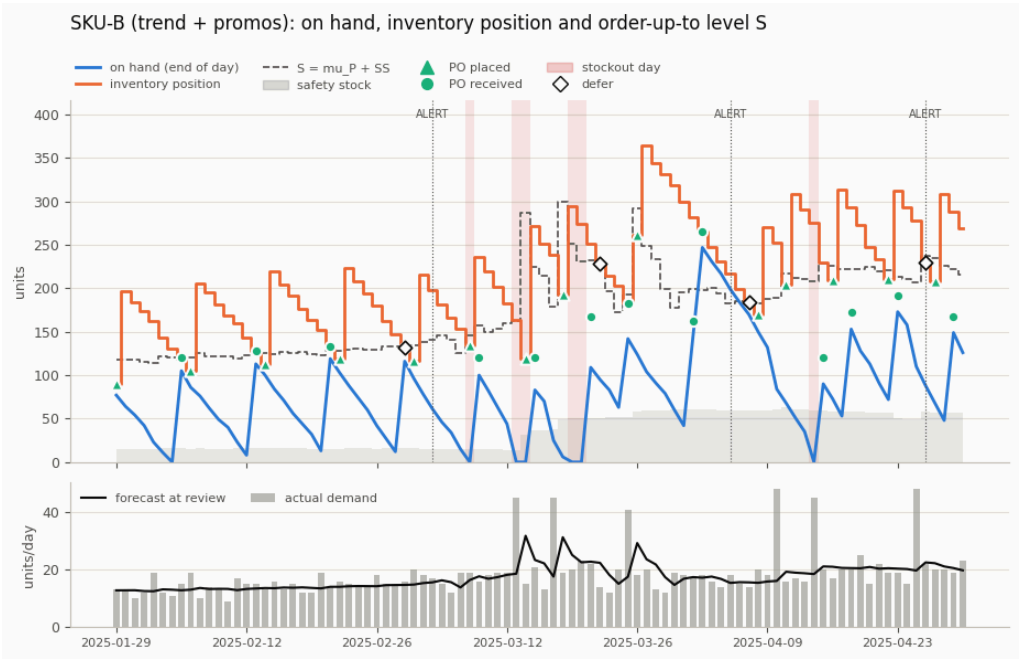
T5. Simulate the horizon: stockouts, fill rate and cost

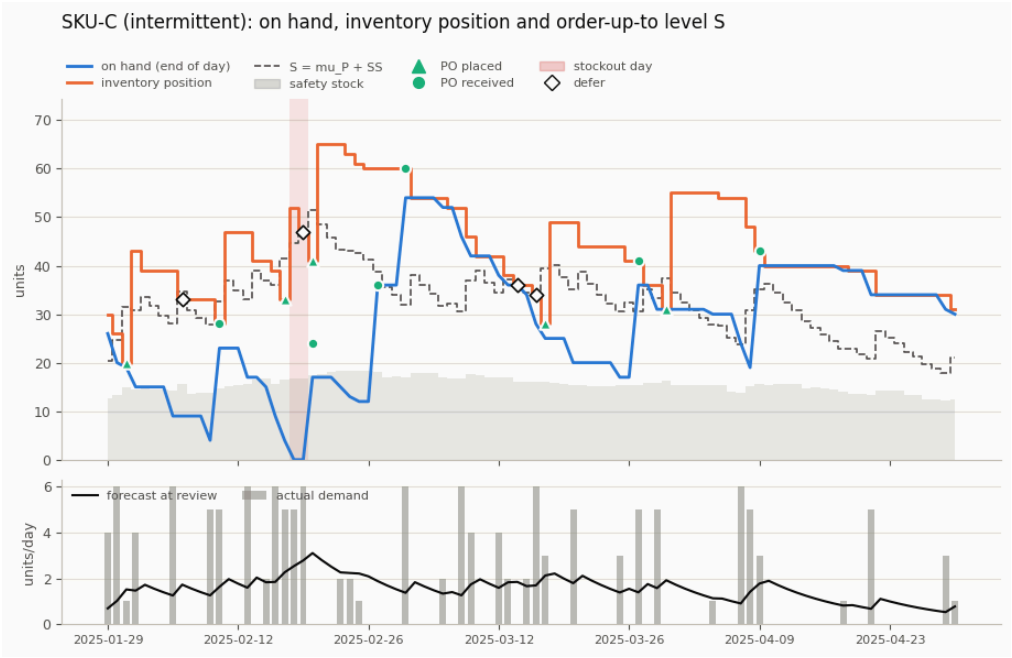
POs arrive exactly L days after they are placed. Costs: holding on end-of-day stock, the stockout cost once per unit short, and the order cost per PO (0 here; the evaluation also runs 25). The charts show on hand, inventory position, S, safety stock, orders placed and received, deferrals, ALERTs and stockout days.

In [8]:
```
table = metrics_table(runs)
cols = ["demand", "fill_rate", "achieved_csl", "stockout_days", "units_short", "pos", "defers", "alerts", "capped",
        "holding_cost", "stockout_cost", "order_cost", "relevant_cost", "purchase_spend", "ending_value",
        "avg_on_hand", "avg_inventory_value", "turns", "forecast_mae", "forecast_bias"]
display(pd.DataFrame(table).T[cols])
for sku, run in runs.items():
    report.plot_inventory_story(run, close=False)
    plt.show()
```

	demand	fill_rate	achieved_csl	stockout_days	units_short	pos	defers	alerts	capped	holding_cost	stockout_cost	order_cost	relevant_cost	purchase_spend	ending_value	avg_on_hand	avg_inventory_value	turns	forecast_mae	forecast_bias
SKU-A	3,649.000	0.999	0.983	1.000	3.000	60.000	7.000	0.000	0.000	108.240	4.500	0.000	112.740	14,200.000	404.000	58.826	235.304	62.030	7.108	0.376
SKU-B	1,692.000	0.960	0.714	6.000	67.000	15.000	4.000	3.000	0.000	727.500	1,675.000	0.000	2,402.500	43,200.000	3,150.000	79.076	1,976.902	21.397	4.499	1.050
SKU-C	144.000	0.951	0.833	2.000	7.000	6.000	4.000	0.000	0.000	1,032.400	280.000	0.000	1,312.400	11,520.000	2,400.000	28.054	2,244.348	5.133	1.926	0.012
TOTAL	5,485.000	0.986	0.924	9.000	77.000	81.000	15.000	3.000	0.000	1,868.140	1,959.500	0.000	3,827.640	68,920.000	5,954.000	165.957	4,456.554	NaN	NaN	NaN


SKU-A (fast mover): on hand, inventory position and order-up-to level S
on hand (end of day)
inventory position
S = mu_P + SS
safety stock
PO placed
PO received
stockout day
defer
units
forecast at review
actual demand
units/day
2025-01-29
2025-02-12
2025-02-26
2025-03-12
2025-03-26
2025-04-09
2025-04-23


SKU-B (trend + promos): on hand, inventory position and order-up-to level S
on hand (end of day)
inventory position
S = mu_P + SS
safety stock
PO placed
PO received
stockout day
defer
ALERT
ALERT
ALERT
units
forecast at review
actual demand
units/day
2025-01-29
2025-02-12
2025-02-26
2025-03-12
2025-03-26
2025-04-09
2025-04-23



T6. Daily agent log

One line per SKU per day. Every rationale is rendered from the numbers in the decision record, so the text and the numbers always agree. The same log is saved to `outputs/agent_log.txt`, and every field is saved to `outputs/decisions.csv`.

In [9]:
```
log = report.write_run_outputs(runs, table, "outputs")
print("\n".join(log))
print()
print("\n".join(report.scorecard(table)))
```

```
d028 2025-01-29 SKU-A  HOLD     0  IP=200 >= S=175.2 (mu_P=141 + SS=34.7) | 5.7 days of cover
d028 2025-01-29 SKU-B  ORDER  120  IP=90 < S=117.9 (mu_P=103 + SS=15.4, z=2.05, sigma_P=7.5) | gap 28, MOQ 120 binds | wait: +10.01u short ($250.30) >= $12.00 holding saved -> order
d028 2025-01-29 SKU-C  HOLD     0  IP=30 >= S=20.4 (mu_P=7.6 + SS=12.7) | 43.3 days of cover
d029 2025-01-30 SKU-A  ORDER   60  IP=160 < S=176.6 (mu_P=142 + SS=34.1, z=1.64, sigma_P=20.7) | gap 17, MOQ 60 binds | wait: +2.29u short ($3.44) >= $1.20 holding saved -> order
d029 2025-01-30 SKU-B  HOLD     0  IP=197 >= S=117.7 (mu_P=103 + SS=15.1) | 15.4 days of cover
d029 2025-01-30 SKU-C  HOLD     0  IP=26 >= S=24.6 (mu_P=11.3 + SS=13.3) | 25.4 days of cover
d030 2025-01-31 SKU-A  ORDER   60  IP=172 < S=181.8 (mu_P=147 + SS=34.4, z=1.64, sigma_P=20.9) | gap 10, MOQ 60 binds | wait: +1.23u short ($1.85) >= $1.20 holding saved -> order
d030 2025-01-31 SKU-B  HOLD     0  IP=184 >= S=117.5 (mu_P=103 + SS=14.7) | 14.3 days of cover
d030 2025-01-31 SKU-C  ORDER   24  IP=20 < S=31.6 (mu_P=16.7 + SS=14.8, z=2.05, sigma_P=7.2) | gap 12, MOQ 24 binds | wait: +0.55u short ($21.85) >= $9.60 holding saved -> order
d031 2025-02-01 SKU-A  HOLD     0  IP=191 >= S=182.9 (mu_P=149 + SS=33.8) | 5.1 days of cover
d031 2025-02-01 SKU-B  HOLD     0  IP=174 >= S=115.3 (mu_P=101 + SS=14.8) | 13.8 days of cover
d031 2025-02-01 SKU-C  HOLD     0  IP=43 >= S=30.7 (mu_P=16.2 + SS=14.5) | 29.3 days of cover
d032 2025-02-02 SKU-A  ORDER   60  IP=150 < S=183.8 (mu_P=151 + SS=33.2, z=1.64, sigma_P=20.2) | gap 34, MOQ 60 binds | wait: +8.24u short ($12.36) >= $1.20 holding saved -> order
d032 2025-02-02 SKU-B  HOLD     0  IP=162 >= S=114.6 (mu_P=100 + SS=14.5) | 13.0 days of cover
d032 2025-02-02 SKU-C  HOLD     0  IP=39 >= S=33.6 (mu_P=18.9 + SS=14.7) | 22.6 days of cover
d033 2025-02-03 SKU-A  DEFER    0  IP=180 < S=180.4 (mu_P=148 + SS=32.9, z=1.64, sigma_P=20.0) | gap 1, MOQ 60 binds | wait: +0.44u short ($0.66) < $1.20 holding saved, CSL_wait 0.948 >= floor 0.930 -> defer
d033 2025-02-03 SKU-B  HOLD     0  IP=143 >= S=121.3 (mu_P=105 + SS=16.1) | 10.9 days of cover
d033 2025-02-03 SKU-C  HOLD     0  IP=39 >= S=31.6 (mu_P=17.1 + SS=14.6) | 25.2 days of cover
d034 2025-02-04 SKU-A  ORDER   60  IP=155 < S=175.9 (mu_P=143 + SS=33.2, z=1.64, sigma_P=20.2) | gap 21, MOQ 60 binds | wait: +3.35u short ($5.03) >= $1.20 holding saved -> order
d034 2025-02-04 SKU-B  HOLD     0  IP=131 >= S=120.1 (mu_P=104 + SS=15.8) | 10.0 days of cover
d034 2025-02-04 SKU-C  HOLD     0  IP=39 >= S=29.8 (mu_P=15.3 + SS=14.4) | 28.0 days of cover
d035 2025-02-05 SKU-A  DEFER    0  IP=176 < S=176.7 (mu_P=144 + SS=32.6, z=1.64, sigma_P=19.8) | gap 1, MOQ 60 binds | wait: +0.45u short ($0.68) < $1.20 holding saved, CSL_wait 0.946 >= floor 0.930 -> defer
d035 2025-02-05 SKU-B  HOLD     0  IP=120 >= S=118.4 (mu_P=103 + SS=15.7) | 9.3 days of cover
d035 2025-02-05 SKU-C  HOLD     0  IP=39 >= S=28.1 (mu_P=13.8 + SS=14.3) | 31.1 days of cover
d036 2025-02-06 SKU-A  ORDER   60  IP=141 < S=175.7 (mu_P=144 + SS=32.1, z=1.64, sigma_P=19.5) | gap 35, MOQ 60 binds | wait: +9.04u short ($13.56) >= $1.20 holding saved -> order
d036 2025-02-06 SKU-B  ORDER  120  IP=105 < S=119.8 (mu_P=104 + SS=15.4, z=2.05, sigma_P=7.5) | gap 15, MOQ 120 binds | wait: +2.65u short ($66.19) >= $12.00 holding saved -> order
d036 2025-02-06 SKU-C  DEFER    0  IP=33 < S=34.6 (mu_P=19 + SS=15.5, z=2.05, sigma_P=7.6) | gap 2, MOQ 24 binds | wait: +0.06u short ($2.33) < $9.60 holding saved, CSL_wait 0.968 >= floor 0.960 -> defer
d037 2025-02-07 SKU-A  ORDER   60  IP=153 < S=181.2 (mu_P=149 + SS=32.7, z=1.64, sigma_P=19.9) | gap 29, MOQ 60 binds | wait: +5.85u short ($8.78) >= $1.20 holding saved -> order
d037 2025-02-07 SKU-B  HOLD     0  IP=206 >= S=125.0 (mu_P=109 + SS=15.8) | 15.1 days of cover
d037 2025-02-07 SKU-C  HOLD     0  IP=33 >= S=30.8 (mu_P=17.1 + SS=13.7) | 21.2 days of cover
d038 2025-02-08 SKU-A  ORDER   60  IP=158 < S=190.0 (mu_P=156 + SS=34.3, z=1.64, sigma_P=20.9) | gap 33, MOQ 60 binds | wait: +7.14u short ($10.72) >= $1.20 holding saved -> order
d038 2025-02-08 SKU-B  HOLD     0  IP=196 >= S=121.5 (mu_P=106 + SS=15.2) | 14.8 days of cover
d038 2025-02-08 SKU-C  HOLD     0  IP=33 >= S=29.2 (mu_P=15.4 + SS=13.8) | 23.5 days of cover
d039 2025-02-09 SKU-A  ORDER   60  IP=177 < S=190.6 (mu_P=157 + SS=34.1, z=1.64, sigma_P=20.7) | gap 14, MOQ 60 binds | wait: +1.76u short ($2.65) >= $1.20 holding saved -> order
d039 2025-02-09 SKU-B  HOLD     0  IP=182 >= S=122.1 (mu_P=107 + SS=15.2) | 13.6 days of cover
d039 2025-02-09 SKU-C  HOLD     0  IP=33 >= S=27.8 (mu_P=13.9 + SS=13.9) | 26.2 days of cover
d040 2025-02-10 SKU-A  HOLD     0  IP=202 >= S=189.0 (mu_P=155 + SS=34.2) | 5.2 days of cover
d040 2025-02-10 SKU-B  HOLD     0  IP=169 >= S=121.8 (mu_P=107 + SS=15.2) | 12.7 days of cover
d040 2025-02-10 SKU-C  ORDER   24  IP=28 < S=32.7 (mu_P=18 + SS=14.7, z=2.05, sigma_P=7.1) | gap 5, MOQ 24 binds | wait: +0.14u short ($5.59) < $9.60 holding saved, but CSL_wait 0.919 < floor 0.960 -> order
d041 2025-02-11 SKU-A  ORDER   60  IP=171 < S=185.7 (mu_P=152 + SS=34, z=1.64, sigma_P=20.6) | gap 15, MOQ 60 binds | wait: +1.95u short ($2.93) >= $1.20 holding saved -> order
d041 2025-02-11 SKU-B  HOLD     0  IP=160 >= S=119.1 (mu_P=103 + SS=16) | 12.4 days of cover
d041 2025-02-11 SKU-C  HOLD     0  IP=47 >= S=36.9 (mu_P=21.7 + SS=15.2) | 23.8 days of cover
d042 2025-02-12 SKU-A  HOLD     0  IP=198 >= S=183.0 (mu_P=150 + SS=33.2) | 5.3 days of cover
d042 2025-02-12 SKU-B  HOLD     0  IP=143 >= S=122.8 (mu_P=106 + SS=16.4) | 10.8 days of cover
d042 2025-02-12 SKU-C  HOLD     0  IP=47 >= S=34.9 (mu_P=19.5 + SS=15.4) | 26.5 days of cover
d043 2025-02-13 SKU-A  ORDER   60  IP=165 < S=181.3 (mu_P=148 + SS=33.3, z=1.64, sigma_P=20.2) | gap 17, MOQ 60 binds | wait: +2.26u short ($3.39) >= $1.20 holding saved -> order
d043 2025-02-13 SKU-B  HOLD     0  IP=128 >= S=123.7 (mu_P=108 + SS=15.9) | 9.5 days of cover
d043 2025-02-13 SKU-C  HOLD     0  IP=47 >= S=33.1 (mu_P=17.6 + SS=15.5) | 29.4 days of cover
d044 2025-02-14 SKU-A  ORDER   60  IP=177 < S=185.3 (mu_P=152 + SS=32.9, z=1.64, sigma_P=20.0) | gap 9, MOQ 60 binds | wait: +1.06u short ($1.59) >= $1.20 holding saved -> order
d044 2025-02-14 SKU-B  ORDER  120  IP=113 < S=124.9 (mu_P=109 + SS=15.9, z=2.05, sigma_P=7.8) | gap 12, MOQ 120 binds | wait: +1.47u short ($36.79) >= $12.00 holding saved -> order
d044 2025-02-14 SKU-C  HOLD     0  IP=41 >= S=38.9 (mu_P=22.4 + SS=16.5) | 20.1 days of cover
d045 2025-02-15 SKU-A  ORDER   60  IP=179 < S=195.5 (mu_P=160 + SS=35.1, z=1.64, sigma_P=21.4) | gap 17, MOQ 60 binds | wait: +2.26u short ($3.38) >= $1.20 holding saved -> order
d045 2025-02-15 SKU-B  HOLD     0  IP=220 >= S=124.4 (mu_P=108 + SS=15.9) | 16.2 days of cover
d045 2025-02-15 SKU-C  HOLD     0  IP=41 >= S=36.9 (mu_P=20.2 + SS=16.7) | 22.4 days of cover
d046 2025-02-16 SKU-A  HOLD     0  IP=204 >= S=193.6 (mu_P=158 + SS=35.3) | 5.2 days of cover
d046 2025-02-16 SKU-B  HOLD     0  IP=204 >= S=126.5 (mu_P=110 + SS=16.1) | 14.8 days of cover
d046 2025-02-16 SKU-C  HOLD     0  IP=39 >= S=36.0 (mu_P=20.4 + SS=15.6) | 21.1 days of cover
d047 2025-02-17 SKU-A  ORDER   60  IP=163 < S=192.7 (mu_P=159 + SS=33.8, z=1.64, sigma_P=20.5) | gap 30, MOQ 60 binds | wait: +6.27u short ($9.40) >= $1.20 holding saved -> order
d047 2025-02-17 SKU-B  HOLD     0  IP=191 >= S=125.8 (mu_P=110 + SS=16.1) | 13.9 days of cover
d047 2025-02-17 SKU-C  ORDER   24  IP=33 < S=41.4 (mu_P=24.9 + SS=16.5, z=2.05, sigma_P=8.0) | gap 9, MOQ 24 binds | wait: +0.35u short ($13.91) >= $9.60 holding saved -> order
d048 2025-02-18 SKU-A  HOLD     0  IP=192 >= S=188.2 (mu_P=155 + SS=32.8) | 4.9 days of cover
d048 2025-02-18 SKU-B  HOLD     0  IP=176 >= S=126.6 (mu_P=111 + SS=15.8) | 12.7 days of cover
d048 2025-02-18 SKU-C  HOLD     0  IP=52 >= S=44.7 (mu_P=27.9 + SS=16.8) | 20.5 days of cover
d049 2025-02-19 SKU-A  ORDER   60  IP=164 < S=183.0 (mu_P=151 + SS=31.9, z=1.64, sigma_P=19.4) | gap 20, MOQ 60 binds | wait: +2.93u short ($4.40) >= $1.20 holding saved -> order
d049 2025-02-19 SKU-B  HOLD     0  IP=164 >= S=124.1 (mu_P=109 + SS=14.8) | 12.0 days of cover
d049 2025-02-19 SKU-C  DEFER    0  IP=47 < S=47.5 (mu_P=30.6 + SS=16.9, z=2.05, sigma_P=8.2) | gap 1, MOQ 24 binds | wait: +0.05u short ($2.09) < $9.60 holding saved, CSL_wait 0.977 >= floor 0.960 -> defer
d050 2025-02-20 SKU-A  ORDER   60  IP=172 < S=189.8 (mu_P=157 + SS=33.1, z=1.64, sigma_P=20.1) | gap 18, MOQ 60 binds | wait: +2.60u short ($3.90) >= $1.20 holding saved -> order
d050 2025-02-20 SKU-B  HOLD     0  IP=152 >= S=122.9 (mu_P=108 + SS=14.9) | 11.3 days of cover
d050 2025-02-20 SKU-C  ORDER   24  IP=41 < S=51.5 (mu_P=34.2 + SS=17.3, z=2.05, sigma_P=8.4) | gap 11, MOQ 24 binds | wait: +0.58u short ($23.07) >= $9.60 holding saved -> order
d051 2025-02-21 SKU-A  ORDER   60  IP=180 < S=195.8 (mu_P=162 + SS=33.9, z=1.64, sigma_P=20.6) | gap 16, MOQ 60 binds | wait: +2.15u short ($3.23) >= $1.20 holding saved -> order
d051 2025-02-21 SKU-B  HOLD     0  IP=133 >= S=128.0 (mu_P=112 + SS=15.6) | 9.5 days of cover
d051 2025-02-21 SKU-C  HOLD     0  IP=65 >= S=48.5 (mu_P=30.8 + SS=17.7) | 23.3 days of cover
d052 2025-02-22 SKU-A  HOLD     0  IP=197 >= S=195.6 (mu_P=163 + SS=32.7) | 4.8 days of cover
d052 2025-02-22 SKU-B  ORDER  120  IP=119 < S=127.8 (mu_P=112 + SS=15.5, z=2.05, sigma_P=7.5) | gap 9, MOQ 120 binds | wait: +0.78u short ($19.52) >= $12.00 holding saved -> order
d052 2025-02-22 SKU-C  HOLD     0  IP=65 >= S=45.7 (mu_P=27.7 + SS=18.1) | 25.8 days of cover
d053 2025-02-23 SKU-A  ORDER   60  IP=152 < S=197.1 (mu_P=165 + SS=32.5, z=1.64, sigma_P=19.8) | gap 46, MOQ 60 binds | wait: +15.40u short ($23.10) >= $1.20 holding saved -> order
d053 2025-02-23 SKU-B  HOLD     0  IP=223 >= S=129.5 (mu_P=114 + SS=15.6) | 15.7 days of cover
d053 2025-02-23 SKU-C  HOLD     0  IP=65 >= S=43.3 (mu_P=24.9 + SS=18.4) | 28.7 days of cover
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d054 2025-02-24 SKU-A  ORDER   60  IP=173 < S=196.2 (mu_P=164 + SS=32.5, z=1.64, sigma_P=19.7) | gap 24, MOQ 60 binds | wait: +4.09u short ($6.13) >= $1.20 holding saved -> order
d054 2025-02-24 SKU-B  HOLD     0  IP=208 >= S=130.2 (mu_P=115 + SS=15.6) | 14.5 days of cover
d054 2025-02-24 SKU-C  HOLD     0  IP=63 >= S=43.0 (mu_P=24.6 + SS=18.3) | 28.1 days of cover
d055 2025-02-25 SKU-A  HOLD     0  IP=204 >= S=189.0 (mu_P=159 + SS=30) | 5.1 days of cover
d055 2025-02-25 SKU-B  HOLD     0  IP=194 >= S=129.9 (mu_P=114 + SS=15.6) | 13.6 days of cover
d055 2025-02-25 SKU-C  HOLD     0  IP=61 >= S=42.6 (mu_P=24.4 + SS=18.3) | 27.5 days of cover
d056 2025-02-26 SKU-A  ORDER   60  IP=168 < S=187.3 (mu_P=157 + SS=29.8, z=1.64, sigma_P=18.1) | gap 20, MOQ 60 binds | wait: +3.15u short ($4.73) >= $1.20 holding saved -> order
d056 2025-02-26 SKU-B  HOLD     0  IP=180 >= S=129.6 (mu_P=114 + SS=15.5) | 12.6 days of cover
d056 2025-02-26 SKU-C  HOLD     0  IP=60 >= S=41.3 (mu_P=23 + SS=18.3) | 28.7 days of cover
d057 2025-02-27 SKU-A  HOLD     0  IP=188 >= S=187.4 (mu_P=158 + SS=29.7) | 4.8 days of cover
d057 2025-02-27 SKU-B  HOLD     0  IP=162 >= S=133.1 (mu_P=117 + SS=16.1) | 11.1 days of cover
d057 2025-02-27 SKU-C  HOLD     0  IP=60 >= S=38.7 (mu_P=20.7 + SS=18) | 31.9 days of cover
d058 2025-02-28 SKU-A  ORDER   60  IP=136 < S=192.5 (mu_P=163 + SS=29.7, z=1.64, sigma_P=18.1) | gap 57, MOQ 60 binds | wait: +25.71u short ($38.56) >= $1.20 holding saved -> order
d058 2025-02-28 SKU-B  HOLD     0  IP=147 >= S=133.4 (mu_P=117 + SS=16.1) | 10.0 days of cover
d058 2025-02-28 SKU-C  HOLD     0  IP=60 >= S=35.6 (mu_P=18.6 + SS=17) | 35.4 days of cover
d059 2025-03-01 SKU-A  ORDER   64  IP=138 < S=201.2 (mu_P=170 + SS=31.5, z=1.64, sigma_P=19.1) | gap 64 | wait: +29.48u short ($44.23) >= $1.28 holding saved -> order
d059 2025-03-01 SKU-B  DEFER    0  IP=132 < S=133.4 (mu_P=118 + SS=15.8, z=2.05, sigma_P=7.7) | gap 2, MOQ 120 binds | wait: +0.09u short ($2.28) < $12.00 holding saved, CSL_wait 0.969 >= floor 0.960 -> defer
d059 2025-03-01 SKU-C  HOLD     0  IP=60 >= S=33.9 (mu_P=16.8 + SS=17.1) | 39.3 days of cover
d060 2025-03-02 SKU-A  ORDER   60  IP=162 < S=200.1 (mu_P=169 + SS=31.4, z=1.64, sigma_P=19.1) | gap 39, MOQ 60 binds | wait: +11.35u short ($17.03) >= $1.20 holding saved -> order
d060 2025-03-02 SKU-B  ORDER  120  IP=116 < S=134.5 (mu_P=119 + SS=15.8, z=2.05, sigma_P=7.7) | gap 19, MOQ 120 binds | wait: +4.44u short ($111.05) >= $12.00 holding saved -> order
d060 2025-03-02 SKU-C  HOLD     0  IP=60 >= S=32.0 (mu_P=15.1 + SS=16.9) | 43.7 days of cover
d061 2025-03-03 SKU-A  ORDER   60  IP=185 < S=197.9 (mu_P=167 + SS=31.2, z=1.64, sigma_P=19.0) | gap 13, MOQ 60 binds | wait: +1.69u short ($2.53) >= $1.20 holding saved -> order
d061 2025-03-03 SKU-B  HOLD     0  IP=216 >= S=138.0 (mu_P=123 + SS=15.2) | 14.1 days of cover
d061 2025-03-03 SKU-C  HOLD     0  IP=54 >= S=38.0 (mu_P=20.2 + SS=17.8) | 29.4 days of cover
d062 2025-03-04 SKU-A  HOLD     0  IP=218 >= S=192.5 (mu_P=161 + SS=31.7) | 5.4 days of cover
d062 2025-03-04 SKU-B  ALERT       TS=+12.49 beyond 3 sd of an unbiased forecast (+/-11.6), forecast running low -> alpha 0.10 -> 0.50
d062 2025-03-04 SKU-B  HOLD     0  IP=198 >= S=140.4 (mu_P=125 + SS=15.5) | 12.7 days of cover
d062 2025-03-04 SKU-C  HOLD     0  IP=54 >= S=36.0 (mu_P=18.2 + SS=17.8) | 32.7 days of cover
d063 2025-03-05 SKU-A  ORDER   60  IP=185 < S=189.8 (mu_P=158 + SS=31.9, z=1.64, sigma_P=19.4) | gap 5, MOQ 60 binds | wait: +0.72u short ($1.08) < $1.20 holding saved, but CSL_wait 0.919 < floor 0.930 -> order
d063 2025-03-05 SKU-B  HOLD     0  IP=181 >= S=145.8 (mu_P=130 + SS=15.4) | 11.1 days of cover
d063 2025-03-05 SKU-C  HOLD     0  IP=54 >= S=34.2 (mu_P=16.4 + SS=17.9) | 36.3 days of cover
d064 2025-03-06 SKU-A  HOLD     0  IP=212 >= S=187.5 (mu_P=155 + SS=32.2) | 5.5 days of cover
d064 2025-03-06 SKU-B  HOLD     0  IP=166 >= S=140.5 (mu_P=125 + SS=15.3) | 10.6 days of cover
d064 2025-03-06 SKU-C  HOLD     0  IP=54 >= S=31.6 (mu_P=14.7 + SS=16.9) | 40.4 days of cover
d065 2025-03-07 SKU-A  ORDER   60  IP=168 < S=188.8 (mu_P=157 + SS=31.5, z=1.64, sigma_P=19.1) | gap 21, MOQ 60 binds | wait: +3.46u short ($5.19) >= $1.20 holding saved -> order
d065 2025-03-07 SKU-B  HOLD     0  IP=154 >= S=125.0 (mu_P=111 + SS=14.4) | 11.1 days of cover
d065 2025-03-07 SKU-C  HOLD     0  IP=52 >= S=32.2 (mu_P=15.4 + SS=16.8) | 37.0 days of cover
d066 2025-03-08 SKU-A  ORDER   60  IP=179 < S=191.3 (mu_P=161 + SS=30, z=1.64, sigma_P=18.3) | gap 13, MOQ 60 binds | wait: +1.60u short ($2.40) >= $1.20 holding saved -> order
d066 2025-03-08 SKU-B  ORDER  120  IP=135 < S=146.2 (mu_P=131 + SS=14.9, z=2.05, sigma_P=7.3) | gap 12, MOQ 120 binds | wait: +1.41u short ($35.35) >= $12.00 holding saved -> order
d066 2025-03-08 SKU-C  HOLD     0  IP=52 >= S=30.7 (mu_P=13.9 + SS=16.8) | 41.1 days of cover
d067 2025-03-09 SKU-A  ORDER   60  IP=189 < S=195.7 (mu_P=165 + SS=30.6, z=1.64, sigma_P=18.6) | gap 7, MOQ 60 binds | wait: +0.88u short ($1.32) >= $1.20 holding saved -> order
d067 2025-03-09 SKU-B  HOLD     0  IP=236 >= S=156.8 (mu_P=142 + SS=15.2) | 13.3 days of cover
d067 2025-03-09 SKU-C  HOLD     0  IP=46 >= S=36.9 (mu_P=19.1 + SS=17.7) | 26.5 days of cover
d068 2025-03-10 SKU-A  HOLD     0  IP=206 >= S=196.3 (mu_P=166 + SS=30.5) | 5.0 days of cover
d068 2025-03-10 SKU-B  HOLD     0  IP=220 >= S=150.1 (mu_P=135 + SS=15.3) | 13.1 days of cover
d068 2025-03-10 SKU-C  HOLD     0  IP=42 >= S=38.9 (mu_P=21.6 + SS=17.3) | 21.4 days of cover
d069 2025-03-11 SKU-A  ORDER   60  IP=184 < S=190.5 (mu_P=158 + SS=32.5, z=1.64, sigma_P=19.7) | gap 7, MOQ 60 binds | wait: +0.87u short ($1.30) >= $1.20 holding saved -> order
d069 2025-03-11 SKU-B  HOLD     0  IP=202 >= S=154.0 (mu_P=139 + SS=14.6) | 11.6 days of cover
d069 2025-03-11 SKU-C  HOLD     0  IP=42 >= S=36.4 (mu_P=19.4 + SS=17) | 23.8 days of cover
d070 2025-03-12 SKU-A  HOLD     0  IP=217 >= S=186.3 (mu_P=153 + SS=33.3) | 5.7 days of cover
d070 2025-03-12 SKU-B  HOLD     0  IP=183 >= S=159.7 (mu_P=146 + SS=14) | 10.0 days of cover
d070 2025-03-12 SKU-C  HOLD     0  IP=42 >= S=34.4 (mu_P=17.5 + SS=16.9) | 26.4 days of cover
d071 2025-03-13 SKU-A  ORDER   60  IP=173 < S=188.7 (mu_P=155 + SS=33.3, z=1.64, sigma_P=20.3) | gap 16, MOQ 60 binds | wait: +2.14u short ($3.21) >= $1.20 holding saved -> order
d071 2025-03-13 SKU-B  HOLD     0  IP=164 >= S=162.7 (mu_P=149 + SS=13.9) | 8.8 days of cover
d071 2025-03-13 SKU-C  HOLD     0  IP=38 >= S=37.2 (mu_P=20.1 + SS=17.1) | 20.7 days of cover
d072 2025-03-14 SKU-A  HOLD     0  IP=197 >= S=186.9 (mu_P=154 + SS=32.7) | 5.1 days of cover
d072 2025-03-14 SKU-B  ORDER  168  IP=119 < S=286.5 (mu_P=254 + SS=32.1, z=2.05, sigma_P=15.6) | gap 168 | wait: +31.80u short ($795.08) >= $16.80 holding saved -> order
d072 2025-03-14 SKU-C  DEFER    0  IP=36 < S=36.4 (mu_P=20.3 + SS=16.1, z=2.05, sigma_P=7.8) | gap 1, MOQ 24 binds | wait: +0.04u short ($1.71) < $9.60 holding saved, CSL_wait 0.977 >= floor 0.960 -> defer
d073 2025-03-15 SKU-A  ORDER   60  IP=150 < S=188.3 (mu_P=158 + SS=30.7, z=1.64, sigma_P=18.7) | gap 39, MOQ 60 binds | wait: +11.68u short ($17.53) >= $1.20 holding saved -> order
d073 2025-03-15 SKU-B  HOLD     0  IP=272 >= S=224.2 (mu_P=187 + SS=37) | 11.6 days of cover
d073 2025-03-15 SKU-C  HOLD     0  IP=36 >= S=34.4 (mu_P=18.3 + SS=16.1) | 21.6 days of cover
d074 2025-03-16 SKU-A  ORDER   60  IP=166 < S=190.1 (mu_P=159 + SS=30.7, z=1.64, sigma_P=18.6) | gap 25, MOQ 60 binds | wait: +4.59u short ($6.89) >= $1.20 holding saved -> order
d074 2025-03-16 SKU-B  HOLD     0  IP=251 >= S=214.6 (mu_P=178 + SS=37) | 11.3 days of cover
d074 2025-03-16 SKU-C  DEFER    0  IP=34 < S=34.7 (mu_P=18.7 + SS=16.1, z=2.05, sigma_P=7.8) | gap 1, MOQ 24 binds | wait: +0.05u short ($1.81) < $9.60 holding saved, CSL_wait 0.975 >= floor 0.960 -> defer
d075 2025-03-17 SKU-A  ORDER   60  IP=183 < S=191.4 (mu_P=161 + SS=30.7, z=1.64, sigma_P=18.7) | gap 9, MOQ 60 binds | wait: +1.06u short ($1.58) >= $1.20 holding saved -> order
d075 2025-03-17 SKU-B  HOLD     0  IP=238 >= S=179.1 (mu_P=141 + SS=38.3) | 13.5 days of cover
d075 2025-03-17 SKU-C  ORDER   24  IP=28 < S=39.5 (mu_P=23.4 + SS=16.1, z=2.05, sigma_P=7.9) | gap 12, MOQ 24 binds | wait: +0.60u short ($24.07) >= $9.60 holding saved -> order
d076 2025-03-18 SKU-A  HOLD     0  IP=209 >= S=188.7 (mu_P=158 + SS=30.5) | 5.3 days of cover
d076 2025-03-18 SKU-B  ORDER  120  IP=193 < S=299.1 (mu_P=250 + SS=48.7, z=2.05, sigma_P=23.7) | gap 107, MOQ 120 binds | wait: +30.06u short ($751.38) >= $12.00 holding saved -> order
d076 2025-03-18 SKU-C  HOLD     0  IP=49 >= S=40.1 (mu_P=24.4 + SS=15.8) | 22.1 days of cover
d077 2025-03-19 SKU-A  ORDER   60  IP=179 < S=184.7 (mu_P=154 + SS=30.3, z=1.64, sigma_P=18.4) | gap 6, MOQ 60 binds | wait: +0.78u short ($1.16) < $1.20 holding saved, but CSL_wait 0.909 < floor 0.930 -> order
d077 2025-03-19 SKU-B  HOLD     0  IP=294 >= S=251.7 (mu_P=201 + SS=50.5) | 11.7 days of cover
d077 2025-03-19 SKU-C  HOLD     0  IP=49 >= S=37.6 (mu_P=21.9 + SS=15.7) | 24.6 days of cover
d078 2025-03-20 SKU-A  HOLD     0  IP=201 >= S=183.1 (mu_P=154 + SS=29) | 5.2 days of cover
d078 2025-03-20 SKU-B  HOLD     0  IP=274 >= S=231.4 (mu_P=181 + SS=50.8) | 12.1 days of cover
d078 2025-03-20 SKU-C  HOLD     0  IP=49 >= S=35.1 (mu_P=19.7 + SS=15.4) | 27.3 days of cover
d079 2025-03-21 SKU-A  ORDER   60  IP=155 < S=185.4 (mu_P=157 + SS=28.3, z=1.64, sigma_P=17.2) | gap 31, MOQ 60 binds | wait: +7.94u short ($11.91) >= $1.20 holding saved -> order
d079 2025-03-21 SKU-B  HOLD     0  IP=251 >= S=232.7 (mu_P=182 + SS=50.4) | 11.0 days of cover
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d079 2025-03-21 SKU-C  HOLD     0  IP=44 >= S=38.7 (mu_P=23.3 + SS=15.4) | 20.8 days of cover
d080 2025-03-22 SKU-A  ORDER   60  IP=156 < S=195.8 (mu_P=165 + SS=30.8, z=1.64, sigma_P=18.7) | gap 40, MOQ 60 binds | wait: +12.68u short ($19.02) >= $1.20 holding saved -> order
d080 2025-03-22 SKU-B  DEFER    0  IP=229 < S=229.6 (mu_P=179 + SS=50.4, z=2.05, sigma_P=24.6) | gap 1, MOQ 120 binds | wait: +0.19u short ($4.68) < $12.00 holding saved, CSL_wait 0.979 >= floor 0.960 -> defer
d080 2025-03-22 SKU-C  HOLD     0  IP=44 >= S=36.2 (mu_P=20.9 + SS=15.2) | 23.1 days of cover
d081 2025-03-23 SKU-A  ORDER   60  IP=174 < S=196.0 (mu_P=165 + SS=30.6, z=1.64, sigma_P=18.6) | gap 22, MOQ 60 binds | wait: +3.88u short ($5.82) >= $1.20 holding saved -> order
d081 2025-03-23 SKU-B  HOLD     0  IP=215 >= S=196.8 (mu_P=146 + SS=51.2) | 11.8 days of cover
d081 2025-03-23 SKU-C  HOLD     0  IP=44 >= S=33.9 (mu_P=18.8 + SS=15.1) | 25.7 days of cover
d082 2025-03-24 SKU-A  HOLD     0  IP=200 >= S=193.3 (mu_P=162 + SS=31) | 4.9 days of cover
d082 2025-03-24 SKU-B  HOLD     0  IP=203 >= S=172.5 (mu_P=121 + SS=51.7) | 13.4 days of cover
d082 2025-03-24 SKU-C  HOLD     0  IP=44 >= S=32.2 (mu_P=17 + SS=15.2) | 28.5 days of cover
d083 2025-03-25 SKU-A  ORDER   60  IP=168 < S=189.5 (mu_P=159 + SS=30.5, z=1.64, sigma_P=18.6) | gap 22, MOQ 60 binds | wait: +3.74u short ($5.60) >= $1.20 holding saved -> order
d083 2025-03-25 SKU-B  ORDER  120  IP=183 < S=192.3 (mu_P=140 + SS=51.9, z=2.05, sigma_P=25.3) | gap 10, MOQ 120 binds | wait: +0.44u short ($10.97) < $12.00 holding saved, but CSL_wait 0.954 < floor 0.960 -> order
d083 2025-03-25 SKU-C  HOLD     0  IP=44 >= S=30.6 (mu_P=15.3 + SS=15.4) | 31.7 days of cover
d084 2025-03-26 SKU-A  HOLD     0  IP=197 >= S=186.4 (mu_P=155 + SS=30.9) | 5.1 days of cover
d084 2025-03-26 SKU-B  ORDER  120  IP=262 < S=292.2 (mu_P=234 + SS=58, z=2.05, sigma_P=28.2) | gap 31, MOQ 120 binds | wait: +2.28u short ($56.91) >= $12.00 holding saved -> order
d084 2025-03-26 SKU-C  HOLD     0  IP=41 >= S=32.5 (mu_P=17 + SS=15.4) | 26.5 days of cover
d085 2025-03-27 SKU-A  ORDER   60  IP=158 < S=186.4 (mu_P=156 + SS=30.9, z=1.64, sigma_P=18.8) | gap 29, MOQ 60 binds | wait: +6.29u short ($9.43) >= $1.20 holding saved -> order
d085 2025-03-27 SKU-B  HOLD     0  IP=364 >= S=248.2 (mu_P=189 + SS=59.1) | 15.4 days of cover
d085 2025-03-27 SKU-C  HOLD     0  IP=41 >= S=30.6 (mu_P=15.3 + SS=15.3) | 29.4 days of cover
d086 2025-03-28 SKU-A  ORDER   60  IP=177 < S=186.3 (mu_P=156 + SS=29.9, z=1.64, sigma_P=18.2) | gap 10, MOQ 60 binds | wait: +1.17u short ($1.75) >= $1.20 holding saved -> order
d086 2025-03-28 SKU-B  HOLD     0  IP=344 >= S=233.8 (mu_P=175 + SS=59.3) | 15.8 days of cover
d086 2025-03-28 SKU-C  HOLD     0  IP=36 >= S=35.1 (mu_P=19.3 + SS=15.8) | 20.5 days of cover
d087 2025-03-29 SKU-A  DEFER    0  IP=187 < S=189.5 (mu_P=161 + SS=28.8, z=1.64, sigma_P=17.5) | gap 3, MOQ 60 binds | wait: +0.51u short ($0.76) < $1.20 holding saved, CSL_wait 0.934 >= floor 0.930 -> defer
d087 2025-03-29 SKU-B  HOLD     0  IP=331 >= S=199.3 (mu_P=139 + SS=60.1) | 19.0 days of cover
d087 2025-03-29 SKU-C  HOLD     0  IP=36 >= S=33.2 (mu_P=17.4 + SS=15.8) | 22.8 days of cover
d088 2025-03-30 SKU-A  ORDER   60  IP=145 < S=190.2 (mu_P=161 + SS=28.7, z=1.64, sigma_P=17.5) | gap 46, MOQ 60 binds | wait: +17.70u short ($26.56) >= $1.20 holding saved -> order
d088 2025-03-30 SKU-B  HOLD     0  IP=319 >= S=178.0 (mu_P=118 + SS=60.3) | 21.7 days of cover
d088 2025-03-30 SKU-C  ORDER   24  IP=31 < S=37.4 (mu_P=21.1 + SS=16.3, z=2.05, sigma_P=7.9) | gap 7, MOQ 24 binds | wait: +0.21u short ($8.54) < $9.60 holding saved, but CSL_wait 0.893 < floor 0.960 -> order
d089 2025-03-31 SKU-A  ORDER   60  IP=162 < S=191.1 (mu_P=163 + SS=28.6, z=1.64, sigma_P=17.4) | gap 30, MOQ 60 binds | wait: +7.17u short ($10.76) >= $1.20 holding saved -> order
d089 2025-03-31 SKU-B  HOLD     0  IP=300 >= S=195.1 (mu_P=135 + SS=60.2) | 17.8 days of cover
d089 2025-03-31 SKU-C  HOLD     0  IP=55 >= S=34.4 (mu_P=19 + SS=15.4) | 31.8 days of cover
d090 2025-04-01 SKU-A  HOLD     0  IP=188 >= S=187.3 (mu_P=160 + SS=27.4) | 4.7 days of cover
d090 2025-04-01 SKU-B  HOLD     0  IP=282 >= S=199.6 (mu_P=139 + SS=60.2) | 16.2 days of cover
d090 2025-04-01 SKU-C  HOLD     0  IP=55 >= S=32.5 (mu_P=17.1 + SS=15.4) | 35.3 days of cover
d091 2025-04-02 SKU-A  ORDER   60  IP=156 < S=184.2 (mu_P=157 + SS=27.5, z=1.64, sigma_P=16.7) | gap 29, MOQ 60 binds | wait: +6.99u short ($10.49) >= $1.20 holding saved -> order
d091 2025-04-02 SKU-B  HOLD     0  IP=265 >= S=197.9 (mu_P=138 + SS=60.2) | 15.4 days of cover
d091 2025-04-02 SKU-C  HOLD     0  IP=55 >= S=30.8 (mu_P=15.4 + SS=15.4) | 39.3 days of cover
d092 2025-04-03 SKU-A  ORDER   60  IP=172 < S=186.0 (mu_P=159 + SS=27.4, z=1.64, sigma_P=16.6) | gap 14, MOQ 60 binds | wait: +1.98u short ($2.97) >= $1.20 holding saved -> order
d092 2025-04-03 SKU-B  HOLD     0  IP=247 >= S=201.0 (mu_P=141 + SS=60.2) | 14.0 days of cover
d092 2025-04-03 SKU-C  HOLD     0  IP=55 >= S=29.2 (mu_P=13.9 + SS=15.3) | 43.6 days of cover
d093 2025-04-04 SKU-A  DEFER    0  IP=186 < S=188.6 (mu_P=161 + SS=27.5, z=1.64, sigma_P=16.7) | gap 3, MOQ 60 binds | wait: +0.50u short ($0.75) < $1.20 holding saved, CSL_wait 0.932 >= floor 0.930 -> defer
d093 2025-04-04 SKU-B  HOLD     0  IP=231 >= S=194.5 (mu_P=134 + SS=60.1) | 13.7 days of cover
d093 2025-04-04 SKU-C  HOLD     0  IP=55 >= S=27.9 (mu_P=12.5 + SS=15.4) | 48.5 days of cover
d094 2025-04-05 SKU-A  ORDER   66  IP=130 < S=196.0 (mu_P=167 + SS=28.5, z=1.64, sigma_P=17.3) | gap 66 | wait: +33.49u short ($50.23) >= $1.32 holding saved -> order
d094 2025-04-05 SKU-B  ALERT       TS=-0.30 within 1 sd (+/-1.8) for 7 days -> alpha back to 0.10
d094 2025-04-05 SKU-B  HOLD     0  IP=217 >= S=183.1 (mu_P=123 + SS=59.9) | 14.1 days of cover
d094 2025-04-05 SKU-C  HOLD     0  IP=54 >= S=27.6 (mu_P=12.3 + SS=15.3) | 48.2 days of cover
d095 2025-04-06 SKU-A  ORDER   60  IP=152 < S=196.2 (mu_P=168 + SS=27.9, z=1.64, sigma_P=17.0) | gap 45, MOQ 60 binds | wait: +17.58u short ($26.36) >= $1.20 holding saved -> order
d095 2025-04-06 SKU-B  HOLD     0  IP=199 >= S=185.2 (mu_P=125 + SS=59.9) | 12.7 days of cover
d095 2025-04-06 SKU-C  HOLD     0  IP=54 >= S=25.2 (mu_P=11.1 + SS=14.1) | 53.5 days of cover
d096 2025-04-07 SKU-A  ORDER   60  IP=173 < S=195.0 (mu_P=167 + SS=28, z=1.64, sigma_P=17.0) | gap 23, MOQ 60 binds | wait: +4.22u short ($6.33) >= $1.20 holding saved -> order
d096 2025-04-07 SKU-B  DEFER    0  IP=184 < S=184.6 (mu_P=125 + SS=59.8, z=2.05, sigma_P=29.1) | gap 1, MOQ 120 binds | wait: +0.20u short ($5.04) < $12.00 holding saved, CSL_wait 0.979 >= floor 0.960 -> defer
d096 2025-04-07 SKU-C  HOLD     0  IP=54 >= S=23.8 (mu_P=10 + SS=13.9) | 59.5 days of cover
d097 2025-04-08 SKU-A  HOLD     0  IP=210 >= S=187.3 (mu_P=160 + SS=27.8) | 5.3 days of cover
d097 2025-04-08 SKU-B  ORDER  120  IP=170 < S=183.3 (mu_P=123 + SS=59.9, z=2.05, sigma_P=29.1) | gap 14, MOQ 120 binds | wait: +0.58u short ($14.39) >= $12.00 holding saved -> order
d097 2025-04-08 SKU-C  HOLD     0  IP=48 >= S=30.7 (mu_P=15.6 + SS=15.1) | 33.9 days of cover
d098 2025-04-09 SKU-A  ORDER   60  IP=176 < S=184.1 (mu_P=157 + SS=26.9, z=1.64, sigma_P=16.4) | gap 9, MOQ 60 binds | wait: +1.02u short ($1.53) >= $1.20 holding saved -> order
d098 2025-04-09 SKU-B  HOLD     0  IP=270 >= S=187.2 (mu_P=127 + SS=60) | 17.0 days of cover
d098 2025-04-09 SKU-C  HOLD     0  IP=43 >= S=35.2 (mu_P=19.5 + SS=15.6) | 24.2 days of cover
d099 2025-04-10 SKU-A  HOLD     0  IP=206 >= S=180.8 (mu_P=153 + SS=27.3) | 5.4 days of cover
d099 2025-04-10 SKU-B  HOLD     0  IP=252 >= S=188.9 (mu_P=129 + SS=60.1) | 15.6 days of cover
d099 2025-04-10 SKU-C  HOLD     0  IP=40 >= S=36.3 (mu_P=20.9 + SS=15.4) | 21.1 days of cover
d100 2025-04-11 SKU-A  ORDER   60  IP=170 < S=179.8 (mu_P=153 + SS=27.3, z=1.64, sigma_P=16.6) | gap 10, MOQ 60 binds | wait: +1.25u short ($1.87) >= $1.20 holding saved -> order
d100 2025-04-11 SKU-B  ORDER  120  IP=204 < S=217.6 (mu_P=154 + SS=63.2, z=2.05, sigma_P=30.8) | gap 14, MOQ 120 binds | wait: +0.61u short ($15.34) >= $12.00 holding saved -> order
d100 2025-04-11 SKU-C  HOLD     0  IP=40 >= S=34.4 (mu_P=18.8 + SS=15.6) | 23.4 days of cover
d101 2025-04-12 SKU-A  DEFER    0  IP=182 < S=183.9 (mu_P=156 + SS=27.5, z=1.64, sigma_P=16.7) | gap 2, MOQ 60 binds | wait: +0.46u short ($0.69) < $1.20 holding saved, CSL_wait 0.937 >= floor 0.930 -> defer
d101 2025-04-12 SKU-B  HOLD     0  IP=308 >= S=212.3 (mu_P=152 + SS=60.6) | 16.2 days of cover
d101 2025-04-12 SKU-C  HOLD     0  IP=40 >= S=32.5 (mu_P=16.9 + SS=15.6) | 26.0 days of cover
d102 2025-04-13 SKU-A  ORDER   60  IP=143 < S=183.7 (mu_P=156 + SS=27.3, z=1.64, sigma_P=16.6) | gap 41, MOQ 60 binds | wait: +15.18u short ($22.77) >= $1.20 holding saved -> order
d102 2025-04-13 SKU-B  HOLD     0  IP=291 >= S=210.7 (mu_P=150 + SS=60.6) | 15.5 days of cover
d102 2025-04-13 SKU-C  HOLD     0  IP=40 >= S=30.9 (mu_P=15.2 + SS=15.7) | 28.9 days of cover
d103 2025-04-14 SKU-A  ORDER   60  IP=166 < S=182.9 (mu_P=156 + SS=27.3, z=1.64, sigma_P=16.6) | gap 17, MOQ 60 binds | wait: +2.67u short ($4.01) >= $1.20 holding saved -> order
d103 2025-04-14 SKU-B  HOLD     0  IP=275 >= S=207.7 (mu_P=148 + SS=59.8) | 14.9 days of cover
d103 2025-04-14 SKU-C  HOLD     0  IP=40 >= S=28.5 (mu_P=13.7 + SS=14.8) | 32.1 days of cover
d104 2025-04-15 SKU-A  ORDER   60  IP=182 < S=184.8 (mu_P=158 + SS=27.2, z=1.64, sigma_P=16.5) | gap 3, MOQ 60 binds | wait: +0.51u short ($0.77) < $1.20 holding saved, but CSL_wait 0.930 < floor 0.930 -> order
d104 2025-04-15 SKU-B  HOLD     0  IP=230 >= S=228.4 (mu_P=169 + SS=59.3) | 10.9 days of cover
d104 2025-04-15 SKU-C  HOLD     0  IP=40 >= S=27.2 (mu_P=12.3 + SS=14.8) | 35.7 days of cover
d105 2025-04-16 SKU-A  HOLD     0  IP=215 >= S=180.3 (mu_P=153 + SS=27.6) | 5.6 days of cover
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d105 2025-04-16 SKU-B  ORDER  120  IP=210 < S=226.0 (mu_P=168 + SS=57.7, z=2.05, sigma_P=28.1) | gap 16, MOQ 120 binds | wait: +0.78u short ($19.42) >= $12.00 holding saved -> order
d105 2025-04-16 SKU-C  HOLD     0  IP=40 >= S=25.7 (mu_P=11.1 + SS=14.6) | 39.7 days of cover
d106 2025-04-17 SKU-A  DEFER    0  IP=179 < S=179.5 (mu_P=152 + SS=27.7, z=1.64, sigma_P=16.8) | gap 1, MOQ 60 binds | wait: +0.38u short ($0.56) < $1.20 holding saved, CSL_wait 0.947 >= floor 0.930 -> defer
d106 2025-04-17 SKU-B  HOLD     0  IP=313 >= S=222.6 (mu_P=165 + SS=57.6) | 15.2 days of cover
d106 2025-04-17 SKU-C  HOLD     0  IP=40 >= S=24.5 (mu_P=10 + SS=14.5) | 44.1 days of cover
d107 2025-04-18 SKU-A  ORDER   60  IP=130 < S=184.3 (mu_P=156 + SS=28.1, z=1.64, sigma_P=17.1) | gap 55, MOQ 60 binds | wait: +25.10u short ($37.65) >= $1.20 holding saved -> order
d107 2025-04-18 SKU-B  HOLD     0  IP=293 >= S=222.1 (mu_P=164 + SS=57.6) | 14.2 days of cover
d107 2025-04-18 SKU-C  HOLD     0  IP=40 >= S=22.9 (mu_P=9 + SS=13.9) | 49.0 days of cover
d108 2025-04-19 SKU-A  ORDER   60  IP=137 < S=188.6 (mu_P=162 + SS=26.8, z=1.64, sigma_P=16.3) | gap 52, MOQ 60 binds | wait: +24.30u short ($36.45) >= $1.20 holding saved -> order
d108 2025-04-19 SKU-B  HOLD     0  IP=273 >= S=221.7 (mu_P=164 + SS=57.6) | 13.3 days of cover
d108 2025-04-19 SKU-C  HOLD     0  IP=39 >= S=22.8 (mu_P=9.2 + SS=13.6) | 46.7 days of cover
d109 2025-04-20 SKU-A  ORDER   60  IP=154 < S=189.6 (mu_P=163 + SS=26.8, z=1.64, sigma_P=16.3) | gap 36, MOQ 60 binds | wait: +11.78u short ($17.67) >= $1.20 holding saved -> order
d109 2025-04-20 SKU-B  HOLD     0  IP=248 >= S=224.8 (mu_P=168 + SS=57.1) | 11.8 days of cover
d109 2025-04-20 SKU-C  HOLD     0  IP=39 >= S=21.7 (mu_P=8.3 + SS=13.5) | 51.9 days of cover
d110 2025-04-21 SKU-A  ORDER   60  IP=180 < S=186.9 (mu_P=160 + SS=26.7, z=1.64, sigma_P=16.3) | gap 7, MOQ 60 binds | wait: +0.87u short ($1.31) >= $1.20 holding saved -> order
d110 2025-04-21 SKU-B  HOLD     0  IP=233 >= S=220.0 (mu_P=163 + SS=57.1) | 11.4 days of cover
d110 2025-04-21 SKU-C  HOLD     0  IP=39 >= S=20.8 (mu_P=7.4 + SS=13.3) | 57.6 days of cover
d111 2025-04-22 SKU-A  HOLD     0  IP=206 >= S=184.2 (mu_P=158 + SS=26.5) | 5.2 days of cover
d111 2025-04-22 SKU-B  ORDER  120  IP=211 < S=221.0 (mu_P=164 + SS=56.9, z=2.05, sigma_P=27.7) | gap 11, MOQ 120 binds | wait: +0.48u short ($11.99) < $12.00 holding saved, but CSL_wait 0.955 < floor 0.960 -> order
d111 2025-04-22 SKU-C  HOLD     0  IP=34 >= S=26.5 (mu_P=12.2 + SS=14.3) | 30.7 days of cover
d112 2025-04-23 SKU-A  ORDER   60  IP=171 < S=182.0 (mu_P=156 + SS=26, z=1.64, sigma_P=15.8) | gap 11, MOQ 60 binds | wait: +1.45u short ($2.17) >= $1.20 holding saved -> order
d112 2025-04-23 SKU-B  HOLD     0  IP=312 >= S=213.7 (mu_P=163 + SS=50.7) | 15.3 days of cover
d112 2025-04-23 SKU-C  HOLD     0  IP=34 >= S=25.2 (mu_P=11 + SS=14.2) | 34.1 days of cover
d113 2025-04-24 SKU-A  HOLD     0  IP=197 >= S=180.2 (mu_P=154 + SS=26.2) | 5.1 days of cover
d113 2025-04-24 SKU-B  HOLD     0  IP=293 >= S=211.1 (mu_P=162 + SS=49.2) | 14.5 days of cover
d113 2025-04-24 SKU-C  HOLD     0  IP=34 >= S=24.0 (mu_P=9.9 + SS=14.1) | 37.9 days of cover
d114 2025-04-25 SKU-A  ORDER   60  IP=150 < S=184.1 (mu_P=157 + SS=26.7, z=1.64, sigma_P=16.2) | gap 35, MOQ 60 binds | wait: +10.76u short ($16.14) >= $1.20 holding saved -> order
d114 2025-04-25 SKU-B  HOLD     0  IP=278 >= S=207.1 (mu_P=158 + SS=49.4) | 14.1 days of cover
d114 2025-04-25 SKU-C  HOLD     0  IP=34 >= S=22.3 (mu_P=8.9 + SS=13.4) | 42.1 days of cover
d115 2025-04-26 SKU-A  ORDER   60  IP=160 < S=188.3 (mu_P=162 + SS=26.7, z=1.64, sigma_P=16.2) | gap 29, MOQ 60 binds | wait: +7.30u short ($10.96) >= $1.20 holding saved -> order
d115 2025-04-26 SKU-B  ALERT       TS=+12.63 beyond 3 sd of an unbiased forecast (+/-11.6), forecast running low -> alpha 0.10 -> 0.50
d115 2025-04-26 SKU-B  DEFER    0  IP=230 < S=237.8 (mu_P=180 + SS=57.5, z=2.05, sigma_P=28.0) | gap 8, MOQ 120 binds | wait: +0.40u short ($10.10) < $12.00 holding saved, CSL_wait 0.962 >= floor 0.960 -> defer
d115 2025-04-26 SKU-C  HOLD     0  IP=34 >= S=21.3 (mu_P=8 + SS=13.3) | 46.7 days of cover
d116 2025-04-27 SKU-A  ORDER   60  IP=182 < S=187.4 (mu_P=161 + SS=26.7, z=1.64, sigma_P=16.2) | gap 6, MOQ 60 binds | wait: +0.72u short ($1.08) < $1.20 holding saved, but CSL_wait 0.906 < floor 0.930 -> order
d116 2025-04-27 SKU-B  ORDER  120  IP=208 < S=235.4 (mu_P=178 + SS=57.2, z=2.05, sigma_P=27.9) | gap 28, MOQ 120 binds | wait: +1.80u short ($45.06) >= $12.00 holding saved -> order
d116 2025-04-27 SKU-C  HOLD     0  IP=34 >= S=19.7 (mu_P=7.2 + SS=12.5) | 51.9 days of cover
d117 2025-04-28 SKU-A  HOLD     0  IP=205 >= S=186.1 (mu_P=159 + SS=26.7) | 5.1 days of cover
d117 2025-04-28 SKU-B  HOLD     0  IP=308 >= S=226.2 (mu_P=169 + SS=57.1) | 14.6 days of cover
d117 2025-04-28 SKU-C  HOLD     0  IP=34 >= S=18.8 (mu_P=6.5 + SS=12.3) | 57.7 days of cover
d118 2025-04-29 SKU-A  DEFER    0  IP=180 < S=181.4 (mu_P=153 + SS=28, z=1.64, sigma_P=17.0) | gap 2, MOQ 60 binds | wait: +0.43u short ($0.65) < $1.20 holding saved, CSL_wait 0.941 >= floor 0.930 -> defer
d118 2025-04-29 SKU-B  HOLD     0  IP=288 >= S=221.6 (mu_P=165 + SS=57.1) | 14.0 days of cover
d118 2025-04-29 SKU-C  HOLD     0  IP=34 >= S=18.0 (mu_P=5.8 + SS=12.1) | 64.1 days of cover
d119 2025-04-30 SKU-A  ORDER   60  IP=142 < S=180.8 (mu_P=153 + SS=27.5, z=1.64, sigma_P=16.7) | gap 39, MOQ 60 binds | wait: +13.63u short ($20.44) >= $1.20 holding saved -> order
d119 2025-04-30 SKU-B  HOLD     0  IP=269 >= S=215.4 (mu_P=158 + SS=57.1) | 13.6 days of cover
d119 2025-04-30 SKU-C  HOLD     0  IP=31 >= S=20.9 (mu_P=8.6 + SS=12.4) | 39.9 days of cover

SKU      fill    CSL so_days  short  POs defer alert   holding  stockout  relevant
-------------------------------------------------------------------------------
SKU-A   0.999  0.983       1      3   60     7     0    108.24      4.50    112.74
SKU-B   0.960  0.714       6     67   15     4     3    727.50   1675.00   2402.50
SKU-C   0.951  0.833       2      7    6     4     0   1032.40    280.00   1312.40
TOTAL   0.986  0.924       9     77   81    15     3   1868.14   1959.50   3827.64
```

Evaluation: baselines, ablations and 200 demand paths

One 92-day path is an anecdote, so every policy also runs on fresh demand paths from the same generator. The policies:

- P0 is the course starter's static (s,Q), run through this simulator with its metrics fixed.
- P1 uses a naive forecast.
- P2 is the textbook EWMA agent, with no defer and no reflect.
- P3 is the full agent.
- P4 knows the future and carries no safety stock. It's a reference point, not an optimum.

Two ablations switch off defer and reflect one at a time. The design was settled on dev seeds 1000-1199; these results use held-out seeds 5000 and up.

In [10]:
```
res = evaluate_all(ds, "outputs", mc_seeds=MC_SEEDS, sweep_seeds=SWEEP_SEEDS)
comp = res["comparison"]
print("Committed CSVs, all SKUs:")
display(comp[comp.sku == "TOTAL"].set_index("policy")[["fill_rate", "stockout_days", "units_short", "holding_cost",
                                                       "stockout_cost", "relevant_cost", "pos"]])
print(f"{MC_SEEDS} demand paths, all SKUs (mean, 5th and 95th percentile):")
display(res["summary"][res["summary"].sku == "TOTAL"].set_index("policy"))
print("Paired comparison of relevant cost on the same paths:")
```

```
display(pd.DataFrame(res["deltas"]))
display(Image("outputs/figures/policy_comparison.png"))
```

```
single path (committed CSVs) ...
Monte Carlo over 200 demand seeds ...
service-level sweep over 100 seeds ...
figures ...
wrote outputs
Committed CSVs, all SKUs:
```

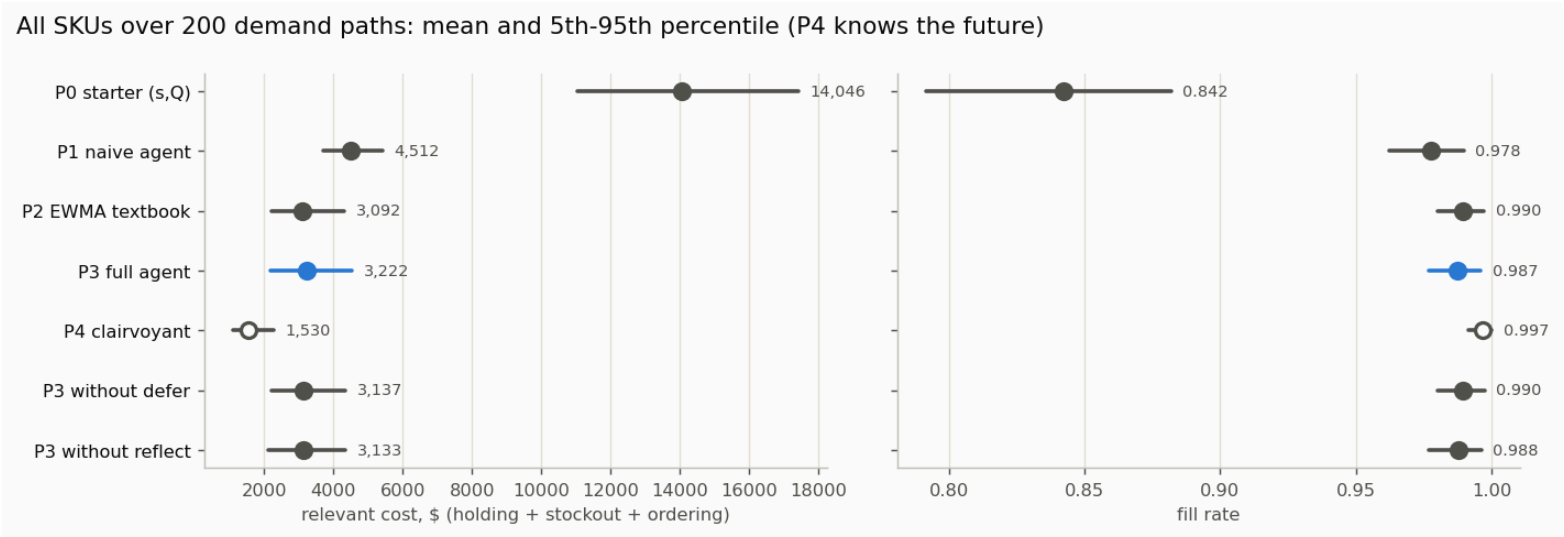
policy	fill_rate	stockout_days	units_short	holding_cost	stockout_cost	relevant_cost	pos
P0 starter (s,Q)	0.810	67	1,043.000	797.220	15,308.500	16,105.720	81
P1 naive agent	0.982	9	100.000	3,787.480	855.000	4,642.480	75
P2 EWMA textbook	0.990	4	53.000	1,906.320	1,325.000	3,231.320	82
P3 full agent	0.986	9	77.000	1,868.140	1,959.500	3,827.640	81
P4 clairvoyant	1.000	0	0.000	1,057.280	0.000	1,057.280	77
P3 without defer	0.990	4	53.000	2,007.120	1,325.000	3,332.120	81
P3 without reflect	0.988	8	67.000	1,802.340	1,709.500	3,511.840	82

```
200 demand paths, all SKUs (mean, 5th and 95th percentile):
```

policy	sku	fill_rate_mean	fill_rate_p5	fill_rate_p95	stockout_days_mean	stockout_days_p5	stockout_days_p95	relevant_cost_mean	relevant_cost_p5	relevant_cost_p95
P0 starter (s,Q)	TOTAL	0.842	0.792	0.882	54.625	45.000	66.000	14,046.475	11,028.542	17,432.612
P1 naive agent	TOTAL	0.978	0.962	0.990	8.555	5.000	12.000	4,512.137	3,711.575	5,415.467
P2 EWMA textbook	TOTAL	0.990	0.980	0.997	5.135	2.000	9.000	3,091.874	2,208.493	4,314.396
P3 full agent	TOTAL	0.987	0.977	0.996	6.270	3.000	10.000	3,221.508	2,175.326	4,528.087
P4 clairvoyant	TOTAL	0.997	0.991	1.000	1.540	0.000	3.000	1,529.578	1,090.701	2,264.391
P3 without defer	TOTAL	0.990	0.980	0.997	5.135	2.000	9.000	3,137.228	2,194.826	4,332.478
P3 without reflect	TOTAL	0.988	0.977	0.996	6.125	3.000	10.000	3,132.851	2,125.886	4,337.502

```
Paired comparison of relevant cost on the same paths:
```

	a	b	metric	mean_delta	p5	p95	share_a_lower	seeds
0	P3 full agent	P2 EWMA textbook	relevant_cost	129.634	-311.840	870.450	0.475	200
1	P3 full agent	P3 without defer	relevant_cost	84.280	-218.953	646.714	0.570	200
2	P3 full agent	P3 without reflect	relevant_cost	88.658	-428.035	650.495	0.330	200
3	P3 full agent	P0 starter (s,Q)	relevant_cost	-10,824.966	-14,364.725	-7,865.709	1.000	200
4	P3 full agent	P1 naive agent	relevant_cost	-1,290.629	-2,040.666	-95.624	0.965	200



In [11]:
```python
# Where each signature feature pays and where it costs, per SKU (P3 minus the ablation, same paths)
mc = res["mc"]
def per_sku(other):
    a = mc[mc.policy == HEADLINE].set_index(["seed", "sku"]).relevant_cost
    b = mc[mc.policy == other].set_index(["seed", "sku"]).relevant_cost
    d = (a - b).groupby("sku")
    return pd.DataFrame({"mean $ difference": d.mean(), "share of paths P3 cheaper": d.apply(lambda s: (s < 0).mean())})
for other in ("P3 without defer", "P3 without reflect", "P2 EWMA textbook"):
    print(f"P3 full agent minus {other}:")
    display(per_sku(other))
```

```
P3 full agent minus P3 without defer:
```

sku	mean $ difference	share of paths P3 cheaper
SKU-A	1.289	0.680
SKU-B	139.319	0.535
SKU-C	-56.328	0.815
TOTAL	84.280	0.570

```
P3 full agent minus P3 without reflect:
```

sku	mean $ difference	share of paths P3 cheaper
SKU-A	0.000	0.000
SKU-B	88.657	0.330
SKU-C	0.000	0.000
TOTAL	88.658	0.330

```
P3 full agent minus P2 EWMA textbook:
```

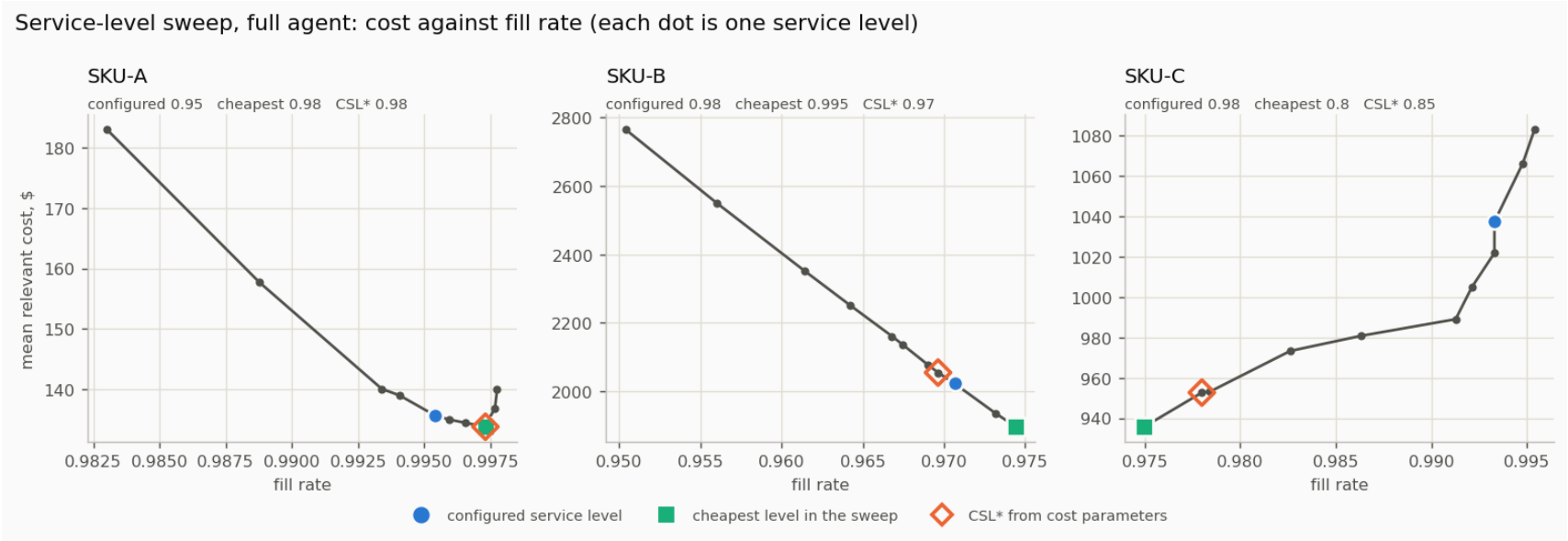
sku	mean $ difference	share of paths P3 cheaper
SKU-A	1.289	0.680
SKU-B	184.673	0.305
SKU-C	-56.328	0.815
TOTAL	129.634	0.475

Service-level sweep

The full agent is run at service levels from 0.80 to 0.995, plus each SKU's CSL*, and mean cost is plotted against fill rate. If CSL* is a good guide, the cheapest level in the sweep should sit near it.

In [12]:
```
sweep = res["sweep"]
best = sweep.loc[sweep.groupby("sku").relevant_cost.idxmin()].set_index("sku")
print(pd.DataFrame({"configured": {s: d["service_level"] for s, d in res["diagnostics"].items()},
                    "CSL*": {s: d["csl_star"] for s, d in res["diagnostics"].items()},
                    "cheapest in sweep": best.service_level}).to_string())
display(Image("outputs/figures/sl_frontier.png"))
```

```
       configured  CSL*  cheapest in sweep
SKU-A       0.950 0.980              0.980
SKU-B       0.980 0.973              0.995
SKU-C       0.980 0.847              0.800
```



In [13]:
```
# Trade-off reflection: inventory vs service
display(Markdown(open("docs/TRADE_OFFS.md", encoding="utf-8").read()))
```

Inventory vs service: the trade-offs

More safety stock buys fewer stockouts with more holding cost, and where the balance sits depends on what a stockout costs compared with a day of holding. The service-level sweep (`outputs/sl_sweep.csv` , 100 held-out demand paths) shows all three cases.

SKU-A is cheap to hold and its stockout cost is comparatively high, so raising its service level from the configured 0.95 to 0.98 lowers mean cost from $135.64 to 133.93$. The configured target is too low for its own economics, and CSL* (0.98) says so.

SKU-C is the opposite. An $80 unit with a 24-unit minimum order sits on the shelf for about two weeks. Holding it at a 0.98 service level costs 1,037 per path, against $936 at 0.80, and the extra spend buys 1.8 points of fill rate. That can still be the right call if missing an order for this item costs more than the 40$ stockout parameter captures, for example by losing a customer. That is a business decision, and the agent leaves it with the buyer.

SKU-B has promo days that a normal model doesn't expect, so even at 0.98 its fill rate is about 0.97. Cost keeps falling all the way to 0.995. The defer rule shows the same tension day to day: on SKU-C, waiting saves holding cost and rarely hurts, while on SKU-B a one-day wait, made while the forecast was running behind the rising trend, produced a 4-unit stockout that cost about eight times the holding it saved.

The cheapest policy isn't automatically the right one. A stockout on a promoted item damages trust in a way a per-unit penalty doesn't measure. That's why the agent reports where cost and service disagree instead of quietly optimizing one of them.

Audit of the course starter notebook

The starter was ported line for line to `baselines/starter_port.py` . Each suspected defect became a property test that runs on both the port (expected to fail, strict xfail) and this agent (must pass). All seven were confirmed; details are in `docs/STARTER_AUDIT.md` . The table below re-runs the starter's own data and parameters, fixing one thing at a time.

In [14]:
```
from baselines.audit import starter_before_after
display(starter_before_after().set_index("version"))
```

version	fill_rate	stockout_days	holding_cost	order_cost	stockout_cost	total_cost
Starter as shipped (its own sim and KPIs)	0.948	NaN	9,301.287	11,000.000	1,858.543	22,159.830
Same policy, metrics fixed (H5 fill rate, H6 stockout charged once)	0.945	32.000	9,301.287	11,000.000	1,550.748	21,852.035
+ protection interval L+1 (H7)	0.979	15.000	10,672.064	11,000.000	582.113	22,254.176
Our textbook agent P2: EWMA, sigma from forecast error (H1-H4, H7), MOQ = EOQ	0.996	3.000	11,752.200	11,000.000	102.238	22,854.438
Our full agent P3: + defer + calibrated reflect, MOQ = EOQ	0.996	3.000	11,623.800	11,000.000	102.238	22,726.038

Tests

The suite covers the hand-worked 6-day trace in `docs/DESIGN.md` , the worked examples from the brief, data loading, and the forecast maths. Hypothesis checks six invariants on 200 random scenarios each: the stock ledger balances, POs arrive exactly L days later, MOQ is respected, costs recompute from the ledger, no decision sees future demand, and the clairvoyant policy never stocks out once its first order can land. The `xfail` lines are the starter defects, failing as expected.

In [15]:
```
out = subprocess.run([sys.executable, "-m", "pytest", "-q", "-rx", "--color=no", "-p", "no:cacheprovider"],
                     capture_output=True, text=True)
print(out.stdout[-6000:])
print("pytest exit code:", out.returncode)
```

```
.....................................x.x.x.x.x.x.x                    [100%]
========================== short test summary info ===========================
XFAIL tests/test_starter_audit.py::test_h1_the_forecast_for_day_t_does_not_see_day_t[starter] - H1: ma7 = rolling(7).mean() includes the same day's demand
XFAIL tests/test_starter_audit.py::test_h2_todays_decision_ignores_demand_after_today[starter] - H2: tool_forecast averages predictions for days i..i+L-1, built from later actuals
XFAIL tests/test_starter_audit.py::test_h3_a_forecastable_series_gets_less_safety_stock[starter] - H3: safety stock uses the std of raw demand, not forecast error
XFAIL tests/test_starter_audit.py::test_h4_a_much_higher_forecast_triggers_an_order[starter] - H4: decide() triggers on the static global ROP; the forecast only sizes the order
XFAIL tests/test_starter_audit.py::test_h5_fill_rate_is_units_filled_over_units_demanded[starter] - H5: kpis() divides by the backlog level summed over days
XFAIL tests/test_starter_audit.py::test_h6_a_unit_short_costs_p_once[starter] - H6: cost_stockout += p * backlog every day the unit stays short
XFAIL tests/test_starter_audit.py::test_h7_known_constant_demand_never_stocks_out_once_orders_can_land[starter] - H7: ROP covers L days, but with daily review the exposure is L+1

pytest exit code: 0
```

In [16]:
```
# Scaling note: deployment, monitoring, reliability, cost controls
display(Markdown(open("docs/SCALING.md", encoding="utf-8").read()))
```

Scaling note

Deployment. Package the `replenish` module in a container and run it once a day as a scheduled job, for example Cloud Scheduler triggering a Cloud Run job or a scheduled ECS task, after the nightly sales export lands. SKUs are independent, so large catalogs shard by SKU across parallel tasks. For the first months the job writes recommended POs for a buyer to approve, not orders straight to the ERP.

Monitoring. Ship the daily log and `decisions.csv` to central logging. Alert when a SKU's rolling 7-day fill rate falls more than 2 points below its target, when the tracking signal raises an ALERT (already in the log), and when forecast MAE drifts above its 28-day baseline. A daily dashboard shows stockouts, fill rate and cost per SKU.

Reliability. Key each run by date and SKU so a retried job can't place a PO twice. Retry ERP and data-feed calls with exponential backoff. Validate inputs before deciding: `SkuParams` already rejects impossible lead times, MOQs and service levels. A missing day in the sales feed must stop the decision for that SKU and raise an alert. Today the code reads a missing day as zero demand, which is right for the order-line CSV but wrong for a feed outage.

Cost controls. The 30-days-of-cover cap on any single PO is already in `plan_order`. `avg_inventory_value` tracks the working capital tied up in stock. In production, add a budget ceiling per category that blocks new POs when stock value would exceed it, and send those blocks to a buyer.